Lab Report 04

Course title: Computer Network Laboratory
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Submitted to-

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Title: DNS Server Configuration

Objective:

- To understand the basic concept of DNS.
- To configure a web server and a DNS server.
- To map a domain name to the IP address of a web server.
- To access a web server using a domain name instead of an IP address.

Description:

In this experiment, one router, two switches, two PCs and two servers were used. Two PCs were connected to the first switch to create one LAN. The two servers were connected to the second switch to create another LAN.

First IP addresses, subnet masks and default gateways were configured for all PCs and servers. The router interfaces were also configured so that both LANs could communicate with each other.

In the second LAN, a server was configured as a web server. From Services - HTTP, the index.html file was edited to create a simple webpage.

Another server was configured as the DNS server. From Services - DNS, a DNS record was added. The name labreport.com was entered and the IP address of the web server was assigned to it. The record was then added to the DNS table.

After configuring the DNS server, its IP address was added as the DNS Server address in the IP configuration of all PCs and the other server. This allowed the devices to use the DNS server for domain name resolution.

Finally, labreport.com was entered in the web browser of the PCs and server. The webpage hosted on the web server was successfully accessed using the domain name.

Connection:

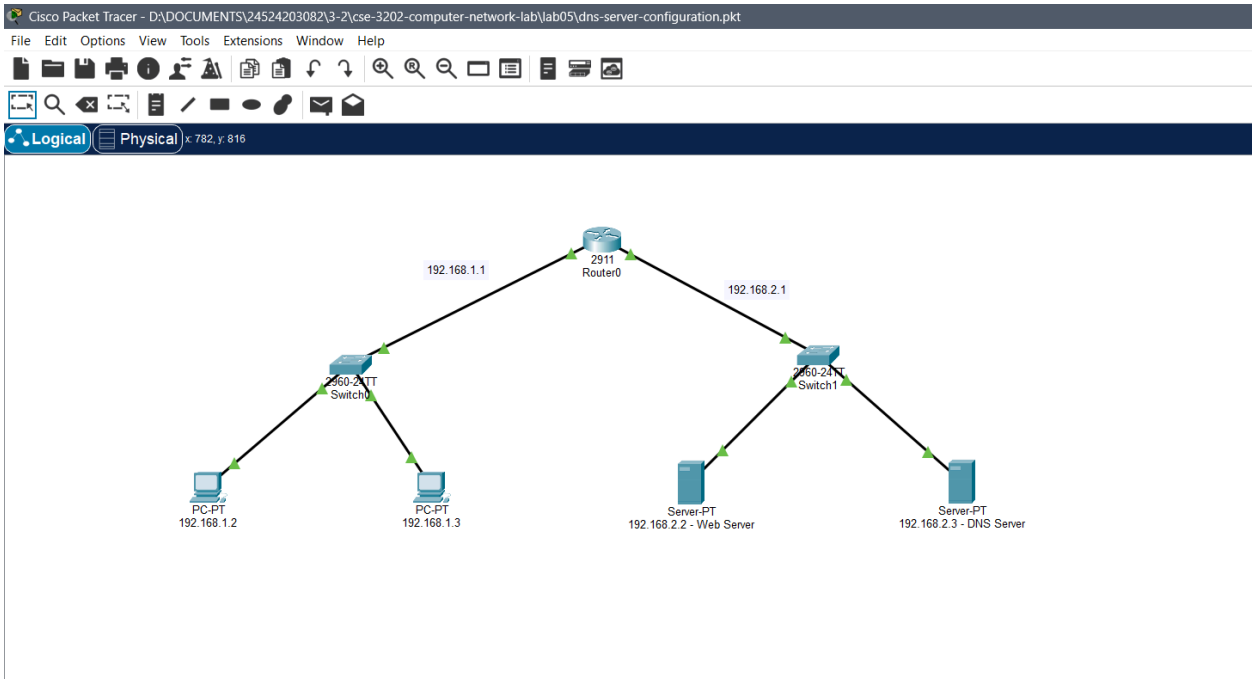


Figure 1: Network Topology for DNS Server Configuration

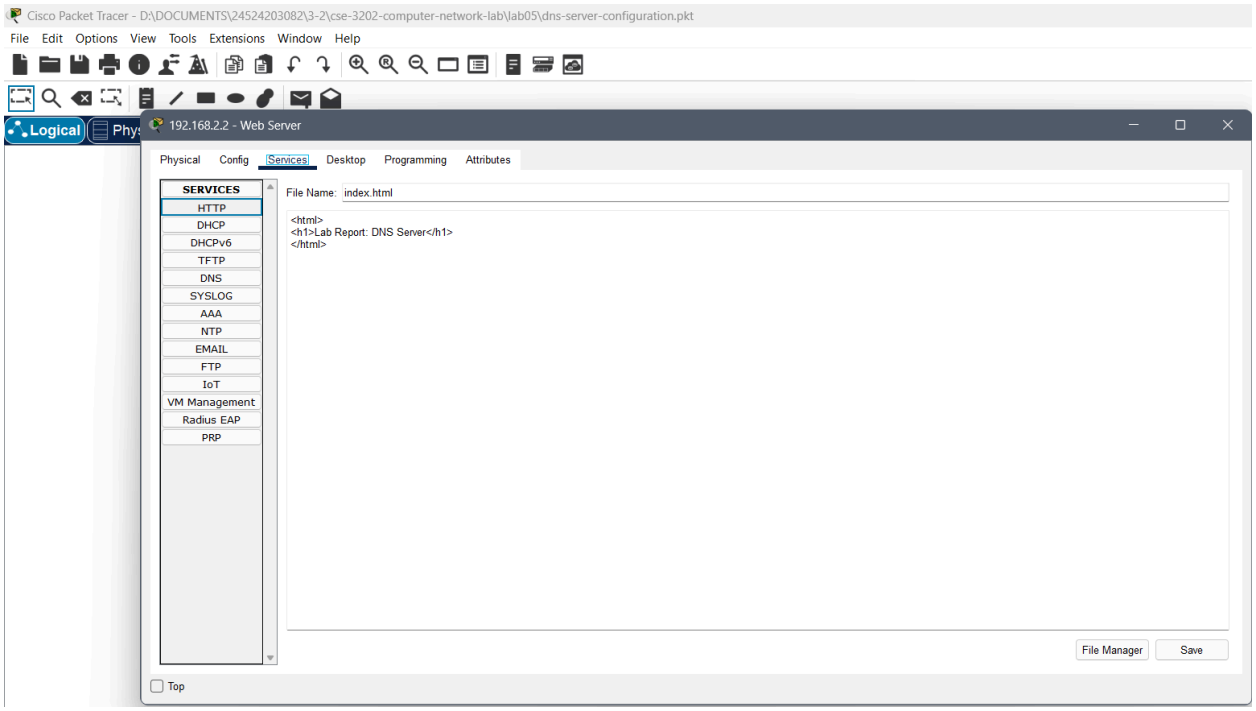


Figure 2: Web Server HTTP Configuration

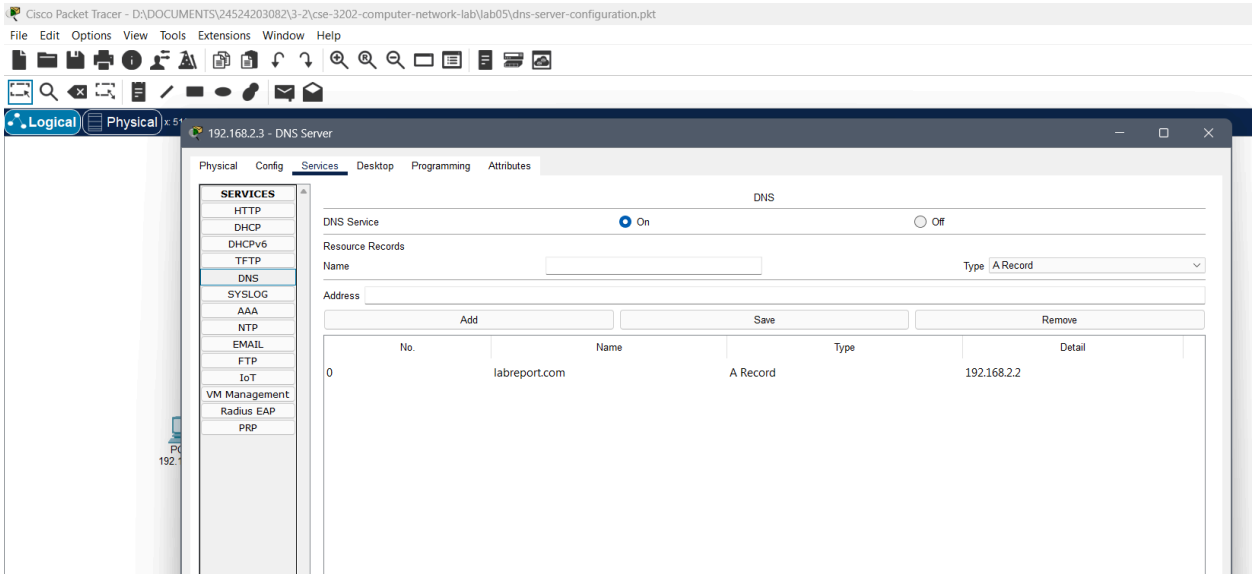


Figure 3: DNS Record for labreport.com

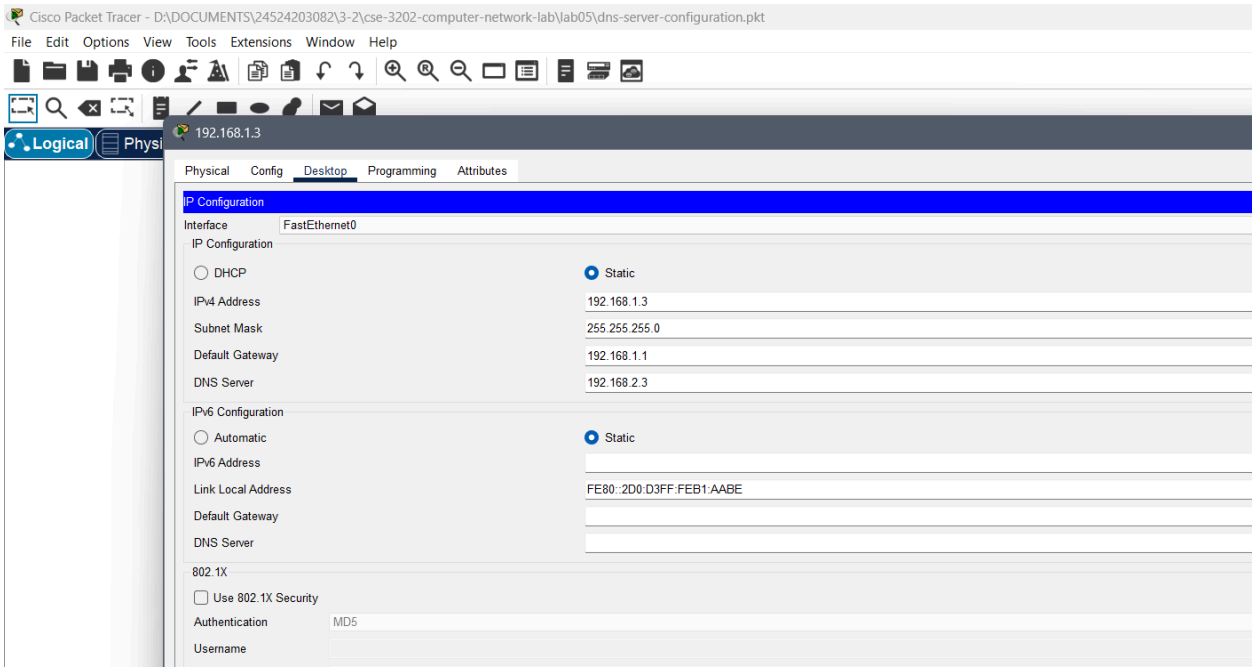


Figure 4: DNS Server IP Added to IP Configuration

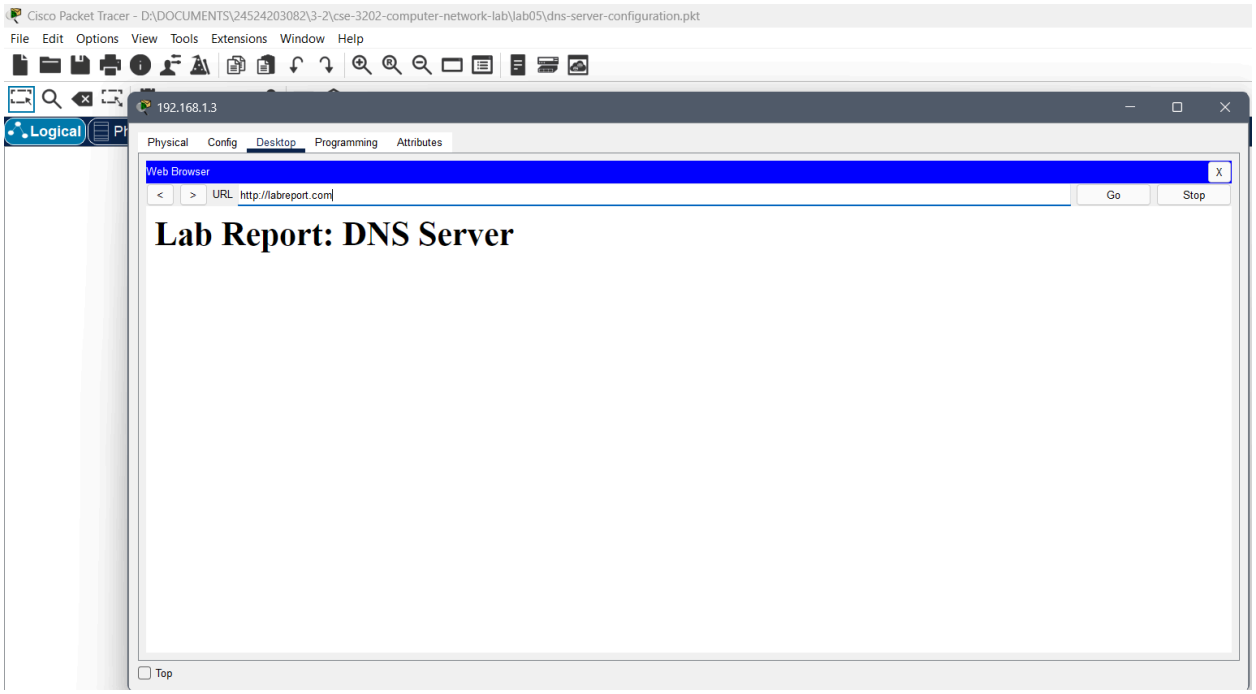


Figure 5: Webpage Accessed Using labreport.com

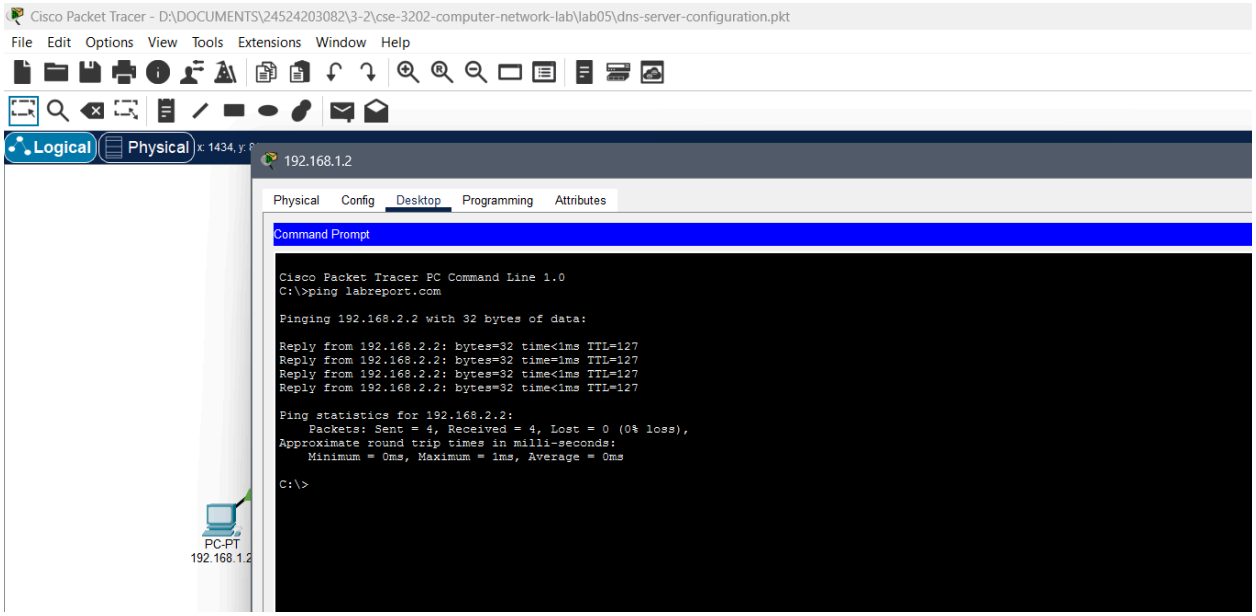


Figure 6: Command Prompt Showing Successful ping labreport.com